

Ayush Sharma

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EDUCATION

New York University

Manhattan, NY

Master of Science in Computer Science | **CGPA: 3.7/4.0**

Aug 2025 – May 2027

Coursework: Big Data, Computer Networks, Operating Systems, Algorithms, Machine Learning, Databases

Vellore Institute of Technology

Vellore, TN

Bachelor of Technology in Computer Science | **CGPA: 3.9/4.0**

Jul 2019 – Jun 2023

Coursework: Data Structures & Algorithms, Artificial Intelligence, Computer Vision, Cybersecurity

TECHNICAL SKILLS, AWARDS, CERTIFICATIONS, OPEN-SOURCE

Languages: Java, Python, C/C++, Go, SQL, MATLAB

Frameworks & Tools: Docker, TensorFlow, PyTorch, REST APIs, Git, Terraform, Linux, Hadoop, Spark, GCP, Azure

Awards: Winner at HackNYU, Startup Week Buildathon, Daytona HackSprint, Elevenlabs Worldwide Hack, Pulse NYC

Certifications: Certified Kubernetes Application Developer (CKAD), AWS Certified Cloud Practitioner

Open-Source: Google Summer of Code (GSoC) 2026 – Ceph, a distributed object, block, and file storage platform

Publication: A Comparative Study on Efficiencies of Variants of CNNs on Image Classification. [\[Paper\]](#)

EXPERIENCE

Software Engineer Intern

Jun 2026 – Present

NYU Center for Urban Science + Progress | Urban Modeling Lab | Advisor: Prof. Debra Laefer

New York, NY

- Architected medallion data lakehouse on Amazon S3 and web-based 3D LiDAR viewer over cube-partitioned Parquet, streaming **10+ TB (75B points) across 3 city-scale surveys** to a custom WebGL renderer.
- Cut read latency 3–4x** over a per-file boto3 pipeline using a reversible spatial hash index that prunes ~70% of files before I/O, PyArrow predicate pushdown with concurrent reads, and a 9-byte/point binary wire format.
- Adapted **Vision Transformers** to 3D point-cloud classification across 11 object classes by restricting self-attention to a 26-neighbour voxel-grid adjacency for local spatial reasoning.

Machine Learning Research Intern

Feb 2026 – April 2026

Courant Institute of Mathematical Sciences | CILVR Lab | Advisor: Prof. Pavel Izmailov

New York, NY

- Built SVG generation model on 89K files achieving **85% XML validity, 78% render rate, 200ms p95 latency**
- Trained Transformers from 1M to 88M params on 100M+ tokens to study scaling laws; implemented μ P zero-shot hyperparameter transfer, **saving 4-5 tuning runs and 10+ GPU hours** per scale

Software Engineer I

Jan 2023 – Jul 2025

Hewlett Packard Enterprise

Bengaluru, KA

- Maintained a carrier-grade media server handling SIP signaling, RTP media, and real-time speech recognition/synthesis for **100M+ users at 99.999% availability**
- Designed & **patented** an on-device LLM inference pipeline on Gemini Nano with context-aware NPU gating for real-time scam detection, **presented at Mobile World Congress 2024**
- Built distroless images eliminating shells, package managers, & unused OS utilities, **cutting image size by 75%**
- Built CI-integrated dependency tracker adopted by dev, QA, & release teams, **reducing open CVEs by 90%**.
- Set up Prometheus and Grafana **monitoring with alerting on 100+ service metrics**
- Built license generation tool with tamper-proof verification & Fluentd event logging, adopted by 8 teams
- Integrated Selenium GUI feature testing into Jenkins CI/CD pipeline, **reducing QA cycle time by 5 days**.

Software Engineer Intern

Oct 2021 – April 2022

Samsung R&D Institute

Bengaluru, KA

- Developed a C++ library that abstracts TCP, UDP, MQTT, gRPC, HTTP & FTP for mesh communication.
- Delivered **10% lower latency and 15% higher throughput** than raw-TCP baseline through multi-threading, async I/O, connection pooling, zero-copy buffers, & Protobuf serialization.

PROJECTS

Social Recommendation Engine | *FastAPI, NetworkX, Stripe, GPT-5, LangChain*

[\[Project\]](#)

- Built a recommendation engine combining collaborative filtering, temporal/spatial signals, & social graph analysis to match 100k+ check-ins with venues & companions. AI agents automate reservations and Stripe payment flows.

LeetCourt | *React, ElevenLabs, OpenRouter (GPT-5), Daytona sandbox, MongoDB*

[\[Project\]](#)

- Led a team & developed a voice-interactive courtroom simulator with AI judge & counsel agents, transforming PDF case files into real-time adversarial debates, scored by a reactive engine that gamifies legal training.

False Markets | *FastAPI, TypeScript, Next.js, Redis, PostgreSQL, CoinGecko, yfinance, Alpaca*

[\[Project\]](#)

- Built evolutionary trading system on 1-min tick data across 25+ crypto assets. Agents encode strategy genomes: RSI/MACD weights, Kelly sizing & trailing stops that evolve on Sharpe ratio, drawdown, and win rate.